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SUPERSEDE
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January 24, 1967 and
Fed. Spec. SS-C-266d
January 15, 1963

FEDERAL SPECIFICATION

CHALK, MARKING, WHITE AND COLORED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for pressed (dustless and molded chalk).

1.2 Classification.

1.2.1 Grades and compositions. The chalk shall be of the following grades and compositions (see 6.2):

Grade A - Pressed (dustless) chalk.
  Composition I - 85 percent calcium carbonate (white).
  Composition II - 45 percent calcium carbonate (white).
  Composition III - 45 percent calcium carbonate (see 1.2.2, and 1.2.3).
Grade B - Molded chalk.
  Composition I - 90 percent calcium sulfate (white or pink).

1.2.2 Colors. Unless otherwise specified, composition III chalk furnished in single colors shall be one of the following colors, as specified (see 6.2): Blue, Green, Orange, Red, or Yellow.

1.2.3 Colored assortment. When specified (see 6.2), composition III chalk shall be furnished in sets of assorted colors, six of each of the following colors:

Black	Brown (dark)	Green (dark)	Red (oxide)
Blue	Burnt Sienna	Magenta	Violet
Blue-Green	Flesh	Orange	White
Blue (dark)	Gold ochre	Red	Yellow
Blue (light)	Gray	Red-orange	Yellow-green
Blue (peacock)	Green	Red-violet	Yellow-orange

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of solicitation for bids or reques. for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7510

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Rate Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

C 25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime.
C 471 - Chemical Analysis of Gypsum and Gypsum Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Toxicity. The chalk shall contain no toxic materials in sufficient quantities to be injurious to the human body.

3.2 Materials. All materials used in the composition of the chalk shall be free from grit, hard spots, and sandy abrasives.

3.2.1 Grade A, pressed (dustless) chalk. Composition I chalk (white) shall contain not less than 85 percent, by weight, pure calcium carbonate. Composition II chalk (white), and composition III chalk (colored), shall contain not less than 45 percent, by weight, pure calcium carbonate, when tested in accordance with 4.4.3. Colored chalk shall contain materials and pigments suitable to produce markings of the desired color.

3.2.2 Grade B, molded chalk. Composition I chalk (white or pink) shall contain not less than 90 percent, by weight, calcium sulfate ($CaSO_4 \cdot 2H_2O$), when tested as specified in 4.4.3.

3.2.3 Other materials. Binding materials and ingredient necessary to control the hardness and dustless characteristics of the chalk shall make up the remainder of the chalk composition.

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3.3 Performance. All chalk shall be capable of producing distinct and uniform markings on standard chalkboards which can be easily erased. Grade A, composition III colored chalk shall also be capable of producing distinct and uniform markings on paper and cloth. The chalk shall be of sufficient hardness to withstand the strength test in 4.4.2 without breaking.

3.4 Shape, size, and weight (grades A and B). The chalk shall be cylindrical in shape and shall be sufficiently straight to pass freely through a tube, when tested as specified in 4.4.1.

3.4.1 Grade A. The chalk shall be 3-1/4 ± 1/16 inches long and 3/8 ± 1/32 inch in diameter. The weight of the chalk shall be not less than 48 ounces per gross.

3.4.2 Grade B. The chalk shall be 3-3/16 ± 1/16 inches long, and 3/8 ± 1/32 inch in diameter, or 7/16 ± 1/32 inch at one end, tapered to 3/8 ± 1/32 inch at the opposite end. The weight of the chalk shall be not less than 24 ounces per gross.

3.5 Workmanship. The chalk shall have a smooth, uniform surface, and shall be free from grit, hard spots, and sandy abrasive, or other defects which may cause undesireable effects on chalkboard surfaces.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Size and lot. All chalk of the same grade, composition, color, or colored assortment presented for inspection at one time shall be considered a lot for purpose of examination and test.

4.2.2 Sampling and inspection for acceptance. Sampling and inspection for acceptance shall be performed in accordance with MIL-STD-105, except where otherwise specified.

4.2.3 Sampling for visual and dimensional inspection. Random sample boxes of chalk shall be taken from each lot in accordance with MIL-STD-105. The sample unit shall be one box of chalk. The chalk shall be examined for defects listed in table I. The Inspection Level shall be S-3, and the Acceptable Quality Level (AQL) shall be 4.0 percent defective.

4.2.4 Sampling for preparation for delivery requirements. Sampling shall be in accordance with MIL-STD-105. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The Inspection level shall be S-2 with an AQL of 6.5 defects per hundred units for the unit container, and S-2 with an AQL of 2.5 defects per hundred units for the shipping container.

4.2.5 Sampling for tests. Random sample boxes of chalk shall be taken from each lot in accordance with MIL-STD-105. The sample unit shall be one box of chalk. The chalk shall be tested as specified in 4.4. The Inspection Level shall be S-2, and the AQL shall be 2.5 percent defective.

4.3 Examination.

4.3.1 Visual and dimensional examination. Samples taken in accordance with 4.2.3 shall be examined for defects listed in table I. Three pieces of chalk, taken at random from each sample box, shall be measured to assure compliance to the size requirements. Any chalk in the sample having one or more defects shall be rejected. If the number of defective units in the sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

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TABLE I. Visual and dimensional inspection

Examine	Defect
Grade and composition	Not as specified (see 1.2.1).
Color(s)	Not as specified (see 1.2.2 and 1.2.3).
Material	Not as specified (see 3.2).
Shape, size and weight	Not as specified (see 3.4, 3.4.1, and 3.4.2).
Workmanship	Not as specified (see 3.5).

4.3.2 Examination of preparation for delivery. Samples selected in accordance with 4.2.4 shall be examined for defects listed in table II, to determine that packaging, packing, and marking requirements conform to the requirements of section 5. Any sample having one or more defects shall be rejected. If the number of defective units in the sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

TABLE II. Inspection for preparation for delivery defects

Examine	Defect
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as improper application of cushioning in unit packages, incomplete closure of container flaps, loose strapping, or distortion of container.
Contents (interior container)	Number of pieces per container is more or less than required.
Contents (shipping container)	Number per container is more or less than required. Net weight exceeds requirements of the box specification.

The common carriers names and addresses should be listed in section 2, where applicable, in accordance with the latest FMS memo.

4.4 Tests. Samples obtained in accordance with 4.2.5 shall be subjected to the following test. Failure of one or more pieces of chalk in a sample box to pass any test shall render the box defective. If the number of defective boxes of chalk exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected. Unless otherwise specified, tests shall be conducted in an atmosphere of 73.5 ± 2°F, and a relative humidity of 50 ± 4 percent. The writing test shall be performed on a broken-in chalkboard. Requirements for breaking-in the chalkboard are as follows:

1. With the side of a piece of chalk, chalk the entire surface.
2. Work the chalk into the writing surface. Use a good quality felt eraser for the purpose.
3. Repeat steps 1 and 2.
4. Clean the chalkboard with a dry soft cloth.

4.4.1 Straightness test. Select 5 pieces of chalk at random from each sample box. Drop each piece of chalk through a hollow vertical tube 5 inches long and 1/16 inch larger in diameter than the maximum diameter of the chalk being tested. Check for compliance with 3.4.

4.4.2 Breaking strength. Select one piece of chalk at random from each sample box. Place chalk perpendicular across two horizontal parallel rods 1/4 inch in diameter and 2-1/2 inches apart. Using a 1/4 inch diameter rod, the test load shall be applied in a vertical direction at the midpoint of the chalk. Gradually apply, and increase test load up to 3 pounds for grade A chalk, and 1-1/2 pounds per grade B chalk, and then maintain maximum test load for 10 seconds. Check for compliance with 3.5.

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4.4.3 Chemical analysis. The calcium carbonate content in grade A chalks shall be tested in accordance with ASTM C25, to determine compliance with 3.2.1. The calcium sulfate content in grade B chalk shall be tested in accordance with the applicable test procedure in ASTM C471, to determine compliance with 3.2.2.

4.4.4 Writing test. Three pieces of each color of chalk shall be taken from each box selected in accordance with 4.2.5. Write ten words on a chalkboard surface and make a few sharp turns, reversals, circles, and 45 degree angles without using excessive pressure. Then erase with a clean felt eraser, and check for compliance with 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. One-hundred and forty-four sticks of white, colored, or colored assortment chalk as specified (see 6.2) shall be packaged to insure protection during shipment and safe delivery to destination. The manufacturers standard package will be acceptable, providing that it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Chalk packaged as specified in 5.1 shall be packed in quantities as specified (see 6.2) in a close-fitting box conforming to PPP-B-636, class-weather resistant, variety DW or to PPP-B-640, class 2. Closure, waterproof sealing and reinforcing of the boxes shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Twelve boxes of chalk packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, variety DW. Closure of the boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The chalk packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to the markings required by 5.3.3 and any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

5.3.3 Special marking. A recognized seal shall be utilized certifying that the contents of the chalk contain no toxic materials in sufficient quantities to be injurious to the human body (see 3.1).

6. NOTES

6.1 Intended use. The chalks covered in this specification are intended primarily for use on standard chalkboards. In addition, the colored chalk, grade A, composition III, may be used on paper and cloth.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade and composition required (see 1.2.).
(c) If grade A, composition III is required specify color desired (see 1.2.3); if colored assortment is required, specify (see 1.2.4).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Any special marking required (see 5.3).

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Preparing activity	User activity:
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